

7 Proving Non-Conditional Statements

Prove the following statements.

2. Suppose $x \in \mathbb{Z}$. Then x is odd if and only if $3x + 6$ is odd.

Proof. First we prove that x being odd implies $3x + 6$ is odd. Suppose x is odd. Then, by definition of an odd number, $x = 2k + 1$ for some integer k . Thus $3x + 6 = 3(2k + 1) + 6 = 6k + 9 = 2(3k + 4) + 1$. Therefore $3x + 6$ is odd by definition of an odd number.

Conversely, we need to prove that $3x + 6$ being odd implies x is odd. We use contrapositive proof. Suppose x is not odd. Then x is even, so $x = 2k$ for some integer k . Thus $3x + 6 = 3(2k) + 6 = 6k + 6 = 2(3k + 3)$. Therefore $3x + 6$ is even by definition of an even number. Thus $3x + 6$ is not odd. Therefore by contrapositive we have proved that $3x + 6$ being odd implies x is odd. $\square$

4. Given an integer a , then $a^3 + a^2 + a$ is even if and only if a is even.

Proof. First we prove that $a^3 + a^2 + a$ being even implies that a is even. We use contrapositive proof. Suppose a is not even, then a is odd. Thus $a = 2k + 1$ for some integer k . Hence, $a^3 + a^2 + a = (2k + 1)^3 + (2k + 1)^2 + 2k + 1$. After expanding we get $16k^3 + 24k^2 + 14k + 2 + 1 = 2(8k^3 + 12k^2 + 7k + 1) + 1$. Therefore, $a^3 + a^2 + a$ is odd by definition of an odd number.

Conversely, we need to prove that a being even implies that $a^3 + a^2 + a$ is even. Suppose a is even. Then $a = 2k$ for some integer k . Then $a^3 + a^2 + a = (2k)^3 + (2k)^2 + (2k) = 8k^3 + 4k^2 + 2k = 2(4k^3 + 2k^2 + k)$. Therefore, $a^3 + a^2 + a$ is even by definition of an even number, $\square$